

Task 1

DNA/Protein Sequence Analysis

1. Introduction

Protein sequence analysis is a fundamental approach in bioinformatics used to characterize the structure, function, and evolutionary relationships of proteins. In this task, **Rubisco (Ribulose biphosphate carboxylase oxygenase)**, the key enzyme responsible for carbon fixation in photosynthesis, was selected from the model plant *Arabidopsis thaliana* (UniProt ID: **O03042**). The primary amino acid sequence was retrieved and analyzed using the Basic Local Alignment Search Tool (BLAST) to identify homologous sequences and analyze evolutionary conservation.

2. Methodology

- **Data Retrieval:** The amino acid sequence of Rubisco from *Arabidopsis thaliana* was retrieved in **FASTA format** from the UniProt database.

The screenshot displays the UniProt database entry for the protein O03042 - RBL_ARATH. The page is organized into a sidebar on the left with navigation links: |Function, Names & Taxonomy, Subcellular Location, Phenotypes & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence, Similar Proteins, and Homologs. The main content area features a header with the protein name and ID, followed by a table of key information: Protein (Ribulose biphosphate carboxylase large chain), Gene (rbcL), Status (UniProtKB reviewed (Swiss-Prot)), Organism (Arabidopsis thaliana (Mouse-ear cress)), Amino acids (479), Protein existence (Evidence at protein level), and Annotation score (50). Below this is a tabbed interface with 'Entry' selected, showing a 'Function' section with a detailed description of the enzyme's role in carbon fixation, a 'Miscellaneous' section describing its tetrameric structure, and a 'Catalytic activity' section with the reaction: 2 (2R)-3-phosphoglycerate + 2 H⁺ = D-ribulose 1,5-bisphosphate + CO₂ + H₂O. The page also includes a 'Tools' section with links for Download, Add, Add a publication, and Entry feedback.

- **Sequence Alignment (BLAST):** The retrieved sequence was submitted to the **NCBI Protein BLAST (blastp)** server. The search was performed against the standard non-redundant (nr) protein database using default parameters (BLOSUM62 matrix, Word size 6, and Existence: 11 Extension: 1 gap costs)

3. Results

The blastp search yielded high-scoring alignments with various plant species. The top hits demonstrated exceptional statistical significance:

Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
972 member(s), 830 organism(s)	green plants	ribulose-1,5-bisphosphate carboxylase/oxygenase large subunit	996	996	100%	0.0	100.00%	479	NP_051067.1
1 member(s), 1 organism(s)	hairy bittercress	unnamed protein product (Cardamine hirsuta)	979	979	100%	0.0	97.69%	658	CAL8659496.1
1 member(s), 1 organism(s)	black tea	unnamed protein product (Camellia sinensis)	953	953	99%	0.0	95.35%	935	CAL5362970.1
1 member(s), 1 organism(s)	black tea	unnamed protein product (Camellia sinensis)	940	940	100%	0.0	94.99%	1053	CAL5362897.1
1 member(s), 1 organism(s)	black tea	unnamed protein product (Camellia sinensis)	939	939	100%	0.0	95.20%	1105	CAL5362979.1
3 member(s), 1 organism(s)	black tea	unnamed protein product (Camellia sinensis)	938	938	100%	0.0	94.99%	1014	CAL5362914.1
1 member(s), 1 organism(s)	black tea	unnamed protein product (Camellia sinensis)	938	938	100%	0.0	95.20%	935	CAL5367250.1
2 member(s), 1 organism(s)	black tea	unnamed protein product (Camellia sinensis)	937	937	100%	0.0	95.20%	969	CAL5362999.1

- **Query Coverage:** 100% for the top homologous sequences, indicating that the alignment spans the entire length of the query protein.
- **E-value (Expect Value):** 0.0 for the leading hits, which strongly proves that the sequence similarity is due to true biological homology and evolutionary conservation rather than random chance.
- **Percentage Identity:** The top results showed high identity percentages (ranging from 95% to 100%), confirming that the Rubisco enzyme is highly conserved across different plant species due to its critical role in photosynthesis.

4. Conclusion

Through this task, successful sequence retrieval and homology analysis were performed. The BLAST results confirm that the Rubisco enzyme maintains a very high degree of sequence conservation throughout plant evolution, highlighting its vital functional importance. This task demonstrates the efficiency of web-based bioinformatics tools like UniProt and NCBI BLAST in analyzing molecular data.